

BLUE RIDGE, VA, USA

WMO: 745985

Lat: 36.631N

Lon: 80.018W

Elev: 287

StdP: 97.93

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 63806

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.7	(c) -7.4	(d) -19.1	(e) 0.7	(f) -4.5	(g) -16.2	(h) 0.9	(i) -2.1	(j) 9.8	(k) 1.1	(l) 8.4	(m) 1.4	(n) 1.0	(o) 320	(p) 0.381

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.5	34.2	23.5	32.9	23.5	32.2	23.3	25.9	31.3	25.2	30.7	24.5	29.6	1.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.7	27.8	23.7	19.3	27.5	22.9	18.2	26.7	81.7	31.4	78.6	30.1	75.8	29.8	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.5	(b) 5.8	(c) 4.9	(e) -14.7	(f) 36.4	(g) 3.8	(h) 1.5	(i) -17.4	(j) 37.5	(k) -19.7	(l) 38.4	(m) -21.8	(n) 39.2	(o) -24.5	(p) 40.3		
(a) 7.5	(b) 5.8	(c) 4.9	(e) -15.2	(f) 27.1	(g) 3.7	(h) 1.1	(i) -17.8	(j) 27.9	(k) -20.0	(l) 28.6	(m) -22.1	(n) 29.2	(o) -24.8	(p) 30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	3.3	5.1	8.9	14.3	18.8	23.0	25.0	24.3	20.6	14.5	8.5	4.7
	DBStd	8.80	5.51	5.05	5.31	4.69	3.96	2.91	2.41	2.63	3.36	4.29	4.78	4.71
	HDD10.0	738	219	150	86	13	1	0	0	0	0	11	83	176
	HDD18.3	2184	467	371	293	136	45	4	0	1	17	132	296	423
	CDD10.0	2311	10	13	53	143	273	392	465	444	318	150	39	11
	CDD18.3	715	0	0	2	15	59	146	207	187	86	13	1	0
	CDH23.3	7164	2	3	50	281	678	1472	1982	1741	756	177	19	3
	CDH26.7	2962	0	0	7	78	228	629	923	786	270	38	2	0
Wind	WSAvg	1.3	1.7	1.8	1.9	1.8	1.3	0.9	0.8	0.7	0.9	1.1	1.4	1.4
Precipitation	PrecAvg	1216	101	73	114	89	107	121	120	115	110	84	85	95
	PrecMax	1680	215	172	227	157	193	226	234	217	310	284	194	202
	PrecMin	914	19	27	17	23	32	24	61	52	20	0	15	38
	PrecStd	206	48	39	52	39	50	56	43	41	71	61	47	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	22.9	27.6	32.3	32.9	36.0	37.0	36.4	33.7	30.0	26.1	22.7
		MCWB	15.3	16.1	16.3	17.9	22.2	23.9	24.4	24.3	22.1	21.2	17.0	15.2
	2%	DB	17.8	19.7	24.1	28.7	31.3	33.1	34.2	34.0	32.1	27.5	22.5	18.6
		MCWB	13.7	12.8	14.6	17.5	21.5	23.7	23.9	24.0	22.8	19.2	15.3	13.6
	5%	DB	15.9	17.4	22.1	27.0	29.1	32.2	32.9	32.6	29.9	25.2	20.2	16.1
		MCWB	11.4	10.9	13.5	16.5	20.6	23.4	24.1	23.7	22.1	17.9	14.4	11.8
	10%	DB	12.6	14.2	18.8	24.0	27.4	30.9	32.0	31.2	27.8	22.8	17.7	13.2
		MCWB	8.7	8.8	12.0	15.5	19.8	22.8	23.9	23.4	21.3	17.0	13.2	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	17.1	18.4	21.5	24.3	25.8	27.3	26.7	25.6	23.3	19.7	17.9
		MCDB	19.4	20.4	23.7	27.3	30.0	31.8	32.9	31.5	29.9	26.5	23.0	19.4
	2%	WB	15.0	14.9	16.8	19.5	22.9	25.0	26.1	25.7	24.3	21.7	17.9	15.3
		MCDB	17.4	18.5	21.8	25.6	28.7	30.8	31.8	31.1	28.3	24.4	20.3	17.9
	5%	WB	12.3	12.3	15.2	18.2	21.8	24.2	25.3	25.0	23.3	19.9	16.0	12.6
		MCDB	14.8	15.3	19.8	23.1	26.9	29.8	30.8	30.4	27.3	23.5	18.9	14.9
	10%	WB	9.1	9.7	13.6	17.0	20.9	23.4	24.6	24.3	22.4	18.3	13.8	10.1
		MCDB	12.1	14.1	17.8	21.8	25.5	28.7	29.5	29.0	26.3	21.4	16.9	12.8
Mean Daily Temperature Range	5% DB	MDBR	12.1	13.0	14.0	14.9	14.2	13.6	12.5	12.8	12.7	14.5	14.3	12.2
		MCDBR	15.9	17.6	18.0	18.9	16.9	15.9	14.8	15.1	16.1	17.8	18.0	16.4
		MCWBR	11.0	11.3	9.7	8.7	8.0	7.1	6.1	6.4	7.7	9.9	11.1	11.0
	5% WB	MCDBR	14.0	14.1	14.4	14.5	14.0	13.6	12.7	13.0	13.0	13.6	13.6	13.1
		MCWBR	11.2	10.7	8.8	7.8	7.1	6.6	5.7	6.0	7.1	8.7	10.2	10.4
Clear-Sky Solar Irradiance	taub	0.303	0.319	0.346	0.393	0.431	0.474	0.509	0.508	0.405	0.360	0.319	0.306	
	taud	2.508	2.453	2.427	2.305	2.217	2.136	2.065	2.043	2.334	2.442	2.516	2.535	
	Ebn at Noon	896	919	921	889	856	816	784	775	848	860	868	869	
	Edh at Noon	80	95	107	128	142	154	165	165	117	96	79	74	
All-Sky Solar Radiation	RadAvg	2.45	3.12	4.17	5.21	5.75	6.22	5.79	5.30	4.42	3.60	2.72	2.09	
	RadStd	0.17	0.38	0.27	0.42	0.48	0.46	0.38	0.28	0.42	0.44	0.22	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-3.04	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (50 neighbors)	+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+139	+88

Nomenclature: See separate page